

Introduction

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Chapter 7: Point Estimation of Parameters

Learning objectives

1. Introduction
2. Sampling Distributions
3. General Concepts Of Point
4. Estimation Methods Of Point Estimation

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Let X is a random variable with probability distribution $f(x)$, which is characterized by the unknown $\theta = (\theta_1, \theta_2, \dots, \theta_k)$

For example, $X \sim N(\mu, \sigma^2)$ then $\theta = (\mu, \sigma^2)$.

How to “determine” the values of θ ?

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In engineering, we often need to estimate

1. The mean μ of a single population.
2. The variance σ^2 (or standard deviation) of a single population.
3. The proportion p of items in a population that belong to a class of interest.
4. The difference in means of two populations, $\mu_1 - \mu_2$.
5. The difference in two population proportions, $p_1 - p_2$.

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The important results on point estimation

1. For μ , the estimate is the $\hat{\mu} = \bar{x}$, **sample mean.**
2. For σ^2 , the estimate is $\hat{\sigma}^2 = s^2$, **the sample variance.**
3. For p , the estimate is $\hat{p} = x/n$, **the sample proportion.**
4. For $\mu_1 - \mu_2$, the estimate is $\hat{\mu}_1 - \hat{\mu}_2 = \bar{x}_1 - \bar{x}_2$
5. For $p_1 - p_2$, the estimate is $\hat{p}_1 - \hat{p}_2$

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Definition

Random Sample

The random variables $X_1, \dots, X_n$ are called a **random sample** of size n if

- The X_i 's are independent
- Every X_i has the same probability distribution

Definition

Statistic

- A **statistic** $\hat{\Theta}$ is any function of the observations $X_1, \dots, X_n$:

$$\hat{\Theta} = h(X_1, \dots, X_n)$$

- The probability distribution of a statistic is called a **sampling distribution**.

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Two important statistic

- Sample mean $\bar{X}$

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}$$

- Sample variance S^2

$$S^2 = \frac{(X_1 - \bar{X})^2 + \dots + (X_n - \bar{X})^2}{n-1}$$

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Theorem

In above example, if $(X_1, \dots, X_n)$ is a random sample of size n take from a normal distribution $N(\mu, \sigma^2)$ then

- $\bar{X}$ has a normal distribution $N(\mu, \sigma^2/n)$

- $\frac{(n-1)S^2}{\sigma^2}$ has a chi-square distribution with $n-1$ degrees of freedom (see pages 273-274).

Theorem

Central Limit Theorem

Let $(X_1, \dots, X_n)$ is a random sample of size n take from a population with mean μ and finite variance σ^2 , and if $\bar{X}$ is the sample mean, the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

As $n \rightarrow \infty$, is the standard normal distribution.

Remark: The normal approximation for $\bar{X}$ depends on the sample size n .

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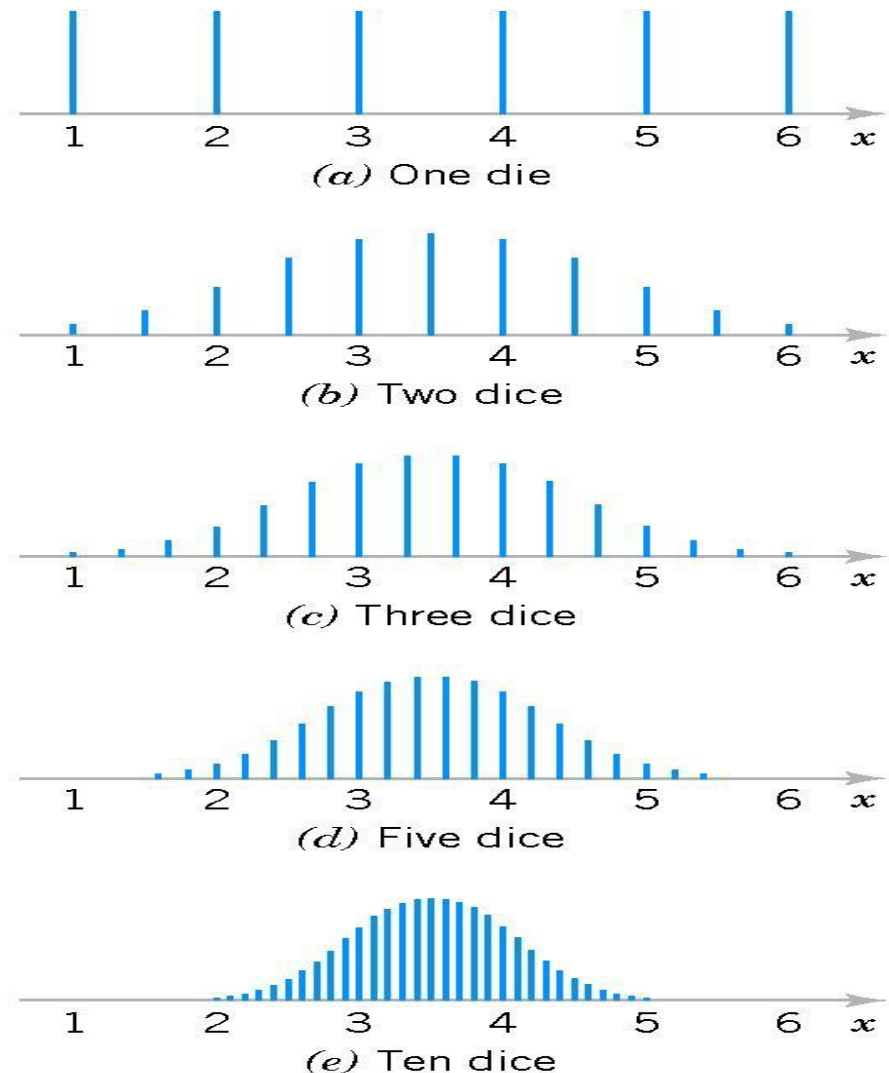
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Figure 7-1
Distributions of average scores from throwing dice.



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Point Estimate

- A **point estimate** of some population parameter θ is a single numerical value $\hat{\theta}$ of a statistic $\hat{\Theta}$.
- The statistic $\hat{\Theta}$ is called the **point estimator**.

Two steps to find point estimation:

Step 1. Determine $\hat{\Theta}$ by using the theoretical results.

Steps 2. Calculate $\hat{\theta}$ from the experimental data.

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Example

Market researchers use the number of sentences per advertisement as a measure of readability for magazine advertisements. The following represents a random sample of the number of sentences found in 15 advertisements. Find a point estimate of the population mean, μ .

9 20 18 16 9 9 11 13 22 16 5 18 6 6 5

$$\text{Step 1. } \hat{\Theta} = \frac{X_1 + \dots + X_{15}}{15}$$

Step 2. A point estimate for μ is

$$\hat{\mu} = \frac{9 + 20 + \dots + 6 + 5}{15} = \frac{183}{15} = 12.2$$

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Unbiased Estimator

The point estimator $\hat{\Theta}$ is an **unbiased estimator** for the parameter θ if

$$E(\hat{\Theta}) = \theta$$

If the estimator is not unbiased, then the difference

$$E(\hat{\Theta} - \theta)$$

is called the **bias** of the estimator $\hat{\Theta}$.

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Example

Suppose that X is a random variable with mean μ and variance σ^2 . Let $X_1, \dots, X_n$ be a random sample of size n from the population represented by X . Show that the sample mean $\bar{X}$ and sample variance S^2 are unbiased estimators of μ and σ^2 , respectively.

For the sample mean.

$$E(\bar{X}) = E \frac{X_1 + \dots + X_n}{n} = \frac{EX_1 + \dots + EX_n}{n} = \mu$$

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Now consider the **sample variance**. We have

$$\begin{aligned}
 E(S^2) &= E \left[\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1} \right] = \frac{1}{n-1} E \sum_{i=1}^n (X_i - \bar{X})^2 \\
 &= \frac{1}{n-1} E \sum_{i=1}^n (X_i^2 + \bar{X}^2 - 2\bar{X}X_i) = \frac{1}{n-1} E \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right) \\
 &= \frac{1}{n-1} \left[\sum_{i=1}^n E(X_i^2) - nE(\bar{X}^2) \right]
 \end{aligned}$$

Hence $E(S^2) = \sigma^2$

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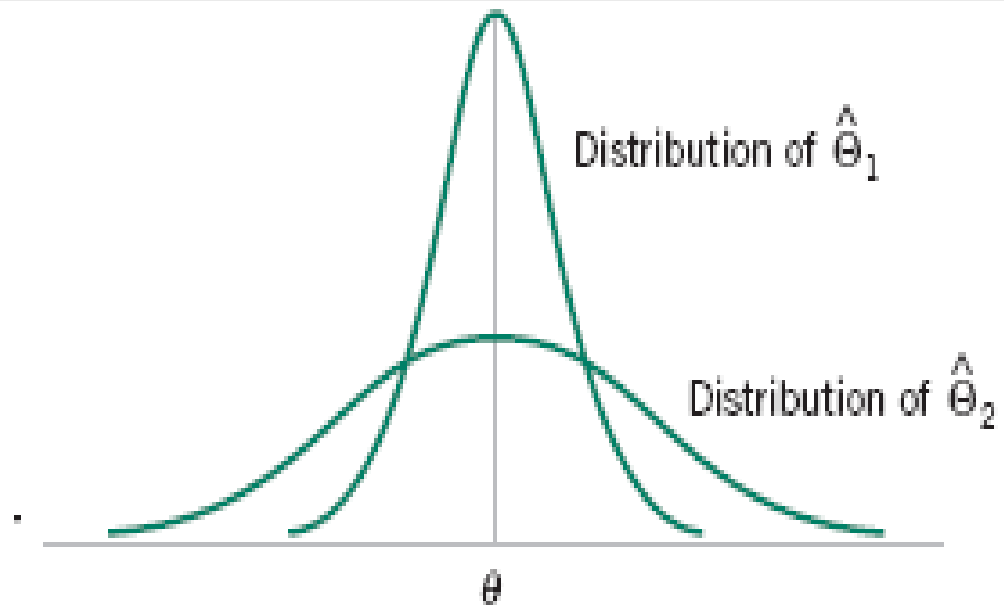
Minimum Variance Unbiased Estimator

If we consider all unbiased estimators of θ , the one with the smallest variance is called the **minimum variance unbiased estimator** (MVUE).

Figure 7-5

The sampling distributions of two unbiased estimators

$\hat{\Theta}_1$ and $\hat{\Theta}_2$.



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Theorem

Let $(X_1, \dots, X_n)$ is a random sample of size n take from a normal distribution $N(\mu, \sigma^2)$, the sample mean $\bar{X}$ is the MVUE for μ .

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Definition Minimum Variance Unbiased Estimator

The **standard error** of an estimator $\hat{\Theta}$ is its standard deviation, given by

$$\sigma_{\hat{\Theta}} = \sqrt{V(\hat{\Theta})}$$

If the standard error involves unknown parameters that can be estimated, substitution of those values into $\sigma_{\hat{\Theta}}$ produces an **estimated standard error**, denoted by $\hat{\sigma}_{\hat{\Theta}}$

Remark

If the random sample of size n take from a normal distribution $N(\mu, \sigma^2)$ then $\bar{X}$ has a normal distribution $N(\mu, \sigma^2/n)$.

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Standard error of $\bar{X}$ is

$$\sigma_{\hat{X}} = \frac{\sigma}{\sqrt{n}}$$

If we did not know σ then the estimated standard error of $\bar{X}$ would be

$$\hat{\sigma}_{\hat{X}} = \frac{s}{\sqrt{n}}$$

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Exercise

A new method of measuring the thermal conductivity of Armco iron: Using a temperature of 100°F and a power input of 550 watts, the following 10 measurements of thermal conductivity (in $\text{Btu/hr-ft-}^{\circ}\text{F}$) were obtained:

41.60, 41.48, 42.34, 41.95, 41.86,
42.18, 41.72, 42.26, 41.81, 42.04

a, Find a point estimate of the mean thermal conductivity at 100°F and 550 watts.

b, Find the estimated standard error of $\bar{X}$

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The k th moments

Let $(X_1, \dots, X_n)$ be a random sample from random variable X with the probability distribution $f(x)$

- The k th moment of X are denoted by $M_k := E(X^k)$
- The corresponding k th **sample moment** is

$$\bar{X}^k = \frac{X_1^k + \dots + X_n^k}{n}$$

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The k th moments

Let $(X_1, \dots, X_n)$ be a random sample from either a probability mass function or probability density function with m unknown parameters $(\theta_1, \theta_2, \dots, \theta_m)$. The moment estimators $\hat{\Theta}_1, \dots, \hat{\Theta}_m$ are found the following system

$$\begin{cases} M_1 = \bar{X}^1 \\ \dots\dots\dots \dots \\ M_m = \bar{X}^m \end{cases}$$

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Example

Suppose that $X_1, \dots, X_n$ be a random sample of size n from a normal distribution $N(\mu, \sigma^2)$. Find the moment estimators of μ and σ^2 .

We have to solve following system

$$\begin{cases} M_1 = \bar{X} \\ M_2 = \bar{X}^2 \end{cases} \quad \text{or} \quad \begin{cases} EX = \frac{X_1 + \dots + X_n}{n} \\ EX^2 = \frac{X_1^2 + \dots + X_n^2}{n} \end{cases}$$

Here $E(X) = \mu$ and $E(X^2) = V(X) + (EX)^2 = \sigma^2 + \mu^2$

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Hence, by solving the above system we have

$$\mu = \bar{X}, \quad \sigma^2 = \frac{\sum_{i=1}^n X_i^2 - n(\bar{X})^2}{n} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n}$$

Finally, the moment estimators are

$$\hat{\mu} = \bar{X},$$

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estimator

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n}$$

Biased
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Maximum likelihood Estimator

Suppose that X is a random variable with probability distribution $f(x, \theta)$, where θ is a single unknown parameter. Let $x_1, x_2, \dots, x_n$ be the observed values in a random sample of size n . Then the **likelihood function** of the sample is

$$L(\theta) = f(x_1, \theta) f(x_2, \theta) \dots f(x_n, \theta)$$

The maximum likelihood estimator (MLE) of θ is the value of θ that maximizes the $L(\theta)$.

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Example

Let X is a Bernoulli random variable. The probability mass function is

$$f(x, p) = \begin{cases} p^x (1-p)^{1-x}, & x = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the maximum likelihood estimator of p .

The likelihood function of a random sample of size n is

$$\begin{aligned} L(p) &= p^{x_1} (1-p)^{1-x_1} p^{x_2} (1-p)^{1-x_2} \cdots p^{x_n} (1-p)^{1-x_n} \\ &= \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i} \end{aligned}$$

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$$\ln L(p) = \left(\sum_{i=1}^n x_i \right) \ln p + \left(n - \sum_{i=1}^n x_i \right) \ln(1 - p)$$

$$\frac{d \ln L(p)}{dp} = \frac{\sum_{i=1}^n x_i}{p} - \frac{\left(n - \sum_{i=1}^n x_i \right)}{1 - p}$$

The maximum likelihood estimator of p is

$$\hat{P} = \frac{X_1 + \dots + X_n}{n}$$

Is this a unbiased estimator?

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Exercise

Let X is a random variable, which has probability mass function is

$$f(x) = \begin{cases} p & , \quad x = -1 \\ 1 - p & , \quad x = 1 \\ 0 & , \quad otherwise \end{cases}$$

Find the maximum likelihood estimator of p .

Solution:

$$f(x) = \begin{cases} p^{\frac{1-x}{2}} (1-p)^{\frac{1+x}{2}} & , \quad x = -1, 1 \\ 0 & , \quad otherwise \end{cases}$$

$$\hat{p} = \frac{1}{2} - \frac{X_1 + \dots + X_n}{2n}$$

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Invariance Property

Let $\hat{\theta}_1, \dots, \hat{\theta}_k$ be the maximum likelihood estimators of the parameters $\theta_1, \dots, \theta_k$. Then the maximum likelihood estimator of any function $h(\theta_1, \dots, \theta_k)$ of these parameters is the same function $h(\hat{\theta}_1, \dots, \hat{\theta}_k)$ of the estimators $\hat{\theta}_1, \dots, \hat{\theta}_k$.

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Exercise

Let X be normally distributed $N(\mu, \sigma^2)$, where both μ and σ^2 are unknown.

- (a) Find the maximum likelihood estimator of μ and σ^2 .
- (b) Find the maximum likelihood estimator of $\theta = \mu^2\sigma^2$.

Solution:

$$\hat{\mu} = \bar{X}, \quad \hat{\sigma}^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n}$$

$$\hat{\theta} = (\bar{X})^2 \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n}$$

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We have studied:

1. Sampling distributions
2. General concepts of point estimation: Unbiased Estimator
3. Methods of point estimation
 - Method of Moments
 - Method of Maximum Likelihood

Homework: Read slides of the next lecture.